HARRISON APPIAH, Ph.D.

Sustainable Chemistry · Waste Valorization · Circular Materials

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RESEARCH PROFILE

Chemist and materials scientist with a Ph.D. in Environmental Science (Material Chemistry) and an M.S. in Biomaterial Science. Research program centers on the chemistry of waste valorization — encompassing solvent-driven chemical recycling of post-consumer plastics, deep-eutectic-solvent (DES) conversion of lignocellulosic biomass to platform chemicals (furfural), biomass pyrolysis and biochar chemistry, and structure–property characterization of waste-derived polymer and biocomposite systems. Combines rigorous analytical and synthetic chemistry with process-scale thinking to advance circular-materials strategies at the interface of sustainable chemical manufacturing, polymer science, and environmental remediation. Author of 15+ peer-reviewed publications and book chapters; industry-ready with hands-on experience in quality-controlled manufacturing environments.

CORE RESEARCH INTERESTS

Waste-derived feedstock chemistry · Chemical & solvent-based plastic recycling · Deep eutectic solvents & green chemistry · Biomass thermochemical conversion · Catalytic/electrified upcycling pathways · Circular polymer systems · Sustainable chemical manufacturing · Advanced materials characterization

RESEARCH EXPERIENCE

Graduate Research & Teaching Assistant

2021 – May 2026

Renewable Materials Laboratory, University of Idaho · Moscow, ID

- Designed and executed multi-phase research on the chemistry of municipal solid waste (MSW) valorization, developing solvent-extraction protocols for selective polymer recovery from heterogeneous waste streams; characterized recovered polymer composition and downstream materials performance via GC-MS, FTIR, UHPLC, and DSC/TGA.
- Investigated deep eutectic solvent (DES) systems for the catalytic depolymerization of Xylan to furfural — a key bio-based platform chemical — correlating DES composition, thermal conditions, and rheological behavior with furfural selectivity and yield; work published in ACS Sustainable Resource Management.
- Conducted pyrolysis/thermochemical conversion of biomass feedstocks to biochar; quantified carbon content, fixed-carbon fraction, nitrogen speciation, and surface area (BET) as functions of processing conditions, informing soil-amendment and carbon-sequestration applications.
- Developed and characterized waste-derived polymer/biocomposite systems using melt extrusion, cast-film processing, and injection molding; established structure–property relationships through DMA, TMA, SEM/AFM, XRD, and rheological profiling.
- Authored and co-authored 15+ peer-reviewed publications, book chapters, and conference presentations spanning plastic recycling chemistry, DES conversion, biomass valorization, and sustainable packaging.
- Mentored undergraduate students in laboratory technique, experimental design, safe chemical handling, and data interpretation; developed instructional materials for undergraduate chemistry laboratory courses.

Laboratory Research Chemist — Biomaterials & Analytical Chemistry

–May 2021

West Virginia University · Morgantown, WV

- Conducted thesis research on xylan valorization from agroforestry waste streams, developing green-chemistry conversion routes to furfural and platform chemicals using DES and hydrothermal processing.
- Designed and validated analytical methods (HPLC, GC, UV-Vis, FTIR) for characterization of lignocellulosic conversion products; maintained research records and contributed experimental support to collaborative project objectives.

Laboratory Technician — Chemistry & Materials Characterization *2021 – 2022*

St. Philip's College · San Antonio, TX

- Operated and maintained advanced analytical instrumentation (SEM, UV-Vis, HPLC, GC, analytical balances); developed and validated analytical methods for chemical sample characterization.
- Prepared reagents, standards, and cultures; supported troubleshooting equipment and laboratory quality standards.

INDUSTRY & APPLIED EXPERIENCE

2026 – Present

Schweitzer Engineering Laboratories (SEL) · Pullman, WA

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EDUCATION

Ph.D., Environmental Science — Material Chemistry *May 2026*

University of Idaho · Moscow, ID · GPA: 3.78/4.0

Dissertation focus: Chemical valorization of municipal solid waste; solvent-based plastic recovery; DES-mediated biomass conversion; waste-derived polymer systems.

M.S., Biomaterial Science (Chemistry) *2021*

West Virginia University · Morgantown, WV · GPA: 3.66/4.0

Thesis: Valorization of xylan in agroforestry waste streams — green-chemistry routes to platform chemicals.

B.S., Chemistry

Kwame Nkrumah University of Science and Technology · Ghana · GPA: 3.1/4.0

Certificate — Semiconductor Fabrication

University of Texas at Austin / Intel Corporation

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- OSHA 30-Hour General Industry Safety & Health — In Progress (2026)
- Semiconductor Fabrication Certificate — University of Texas at Austin / Intel Corporation
- Advanced Laboratory Safety & Chemical Hygiene — University of Idaho
- Environmental Regulations & Compliance for Chemists (pursuing: CHMM preparation) In Progress
- ACS Member — American Chemical Society; Division of Industrial & Engineering Chemistry

SELECTED TECHNICAL SKILLS

Analytical Chemistry GC, GC-MS, Py-GC/MS, UHPLC/HPLC, ICP-MS, UV-Vis, FTIR, XPS

Materials Characterization	DSC, TGA, DMA, TMA, SEM, AFM, XRD, BET, Rheology, Density/Pycnometry
Synthesis & Processing	Deep eutectic solvent synthesis, biomass thermochemical conversion (pyrolysis), melt extrusion, cast-film processing, injection molding, polymer composite fabrication
Research Methods	Experimental design, method development & validation, process optimization, statistical data analysis, scientific writing, QC/QA
Safety & Compliance	Chemical hygiene & hazard communication, OSHA 30-hr GI (in progress), ESD/Lean/5S, PSM fundamentals, laboratory waste management
Software & Data	Python (data analysis), R, OriginPro, ChemDraw, Aspen (familiarity), Microsoft Office Suite

SELECTED PUBLICATIONS & RESEARCH OUTPUTS

Peer-Reviewed Journal Articles

Appiah, H., Asamoah, P., & McDonald, A. (2026). Waste-to-energy technologies and their role in municipal solid waste management. Recycling, 11(3), 56.

Appiah, H., Bar-Ziv, E., Klinger, J., & McDonald, A. (2026). Plastic recovery from municipal solid waste by solvent extraction. ACS Sustainable Resource Management, 3(2), 524–533.

Appiah, H., Bar-Ziv, E., Klinger, J., & McDonald, A. (2025). Exploring new applications of municipal solid waste. Sustainability, 17(8), 3719.

Appiah, H., Dawson-Andoh, B., Oginni, O., & McDonald, A. (2025). Deep eutectic solvents for xylan-based furfural synthesis. ACS Sustainable Resource Management, 2(3), 536–545.

Sánchez-Rivera, K., Granger, C., Appiah, H., et al. (2024). Cast film production with polyethylene recycled from post-industrial printed multilayer film by STRAP. ACS Materials Letters, 6(9), 4042–4050.

Struhs, E., Mirkouei, A., Appiah, H., & McDonald, A. (2024). Catalytic fast pyrolysis and liquid fractionation utilizing a free-fall reactor. Frontiers in Industrial Microbiology, 2, 1426067.

Appiah, H., Park, S. H., & Tongco, J. V. (2026). Green pretreatment, thermochemical conversion, and by-product valorization of lignocellulosic biomass. Journal of Carbon Research.

Yusuf, S., Ewurum, N., Appiah, H., & Tongco, J. (2025). Industrial hemp as precursor for sustainable bioproducts. Fibers, 13(11), 155.

Book Chapters

Appiah, H., Ewurum, N., Yusuf, S., & Tongco, J. (2025). Smart food packaging and solutions. In Polymers and Composite Materials for Packaging (pp. 395–413). Springer Nature.

Appiah, H., Tongco, J. V., Yusuf, S. B., & Marquez, K. P. (2026). Engineering and development of high-entropy alloys. In Multi-scale Mechanics of High Entropy Alloys (pp. 1–23). Springer Nature.

Huber, G., et al. incl. Appiah, H. (2026). Upcycling municipal solid waste to polymers and bioethanol. [Technical Report / Collaborative Publication]

POSTDOCTORAL RESEARCH VISION

Seeking to develop next-generation chemical-conversion strategies for heterogeneous waste-derived feedstocks — with particular focus on (1) electrified and catalytic deconstruction of mixed plastic waste to monomers and chemical intermediates, (2) DES-mediated lignocellulosic fractionation to high-value platform chemicals, and (3) integrated process design coupling waste-stream chemistry with downstream polymer upcycling. Open to industry R&D positions in sustainable chemistry, chemical recycling, and circular materials manufacturing.